
Data Analysis Internship

TASK 2 REPORT

**Sales Data Cleaning, Validation
and Performance Dashboard**

Analysis carried out on

THE FACTSALE DATASET

26,397 order lines, January 2013 to May 2016

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September 8, 2026

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1 Executive summary

The brief anticipates a dirty file: impossible calendar dates, duplicate rows, broken arithmetic. Each of those was tested explicitly. **Most of them are not present.** Every arithmetic identity in the file holds to within one cent, there are no duplicate records, and every date parses without exception.

Reporting corrections that were not needed would have been the easy path. This report does the opposite: it states what the file actually contains, and the three findings that matter are structural rather than arithmetic.

1. **9,077 order lines, 34.4% of the file, carry no customer key.** Every customer metric therefore describes two thirds of the business.
2. **The last year is incomplete,** January to May 2016 only. Any year-on-year chart built without noticing shows a collapse that never happened.
3. **A routine outlier screen would have deleted 48.0% of all revenue.** Sales value is right-skewed by construction, and a symmetric rule applied to a skewed variable removes the top of the distribution rather than errors.

No row was deleted. All 26,397 lines are retained.

2 Key performance indicators

Table 1. The seven indicators selected for the dashboard

KPI	Value	Why it was selected
Total sales, excl. tax	19.88M	The top line every other figure divides into
Gross profit	9.92M	Revenue that does not convert is not performance
Gross margin	49.9%	The one ratio that survives a change of scale
Orders (invoices)	8,188	Separates growth in volume from growth in basket
Average order value	2,428	The lever a commercial team can actually pull
Units sold	1,028,670	Detects margin bought with discount
Identified customers	48	Bounds every customer conclusion in this report

3 The dashboard

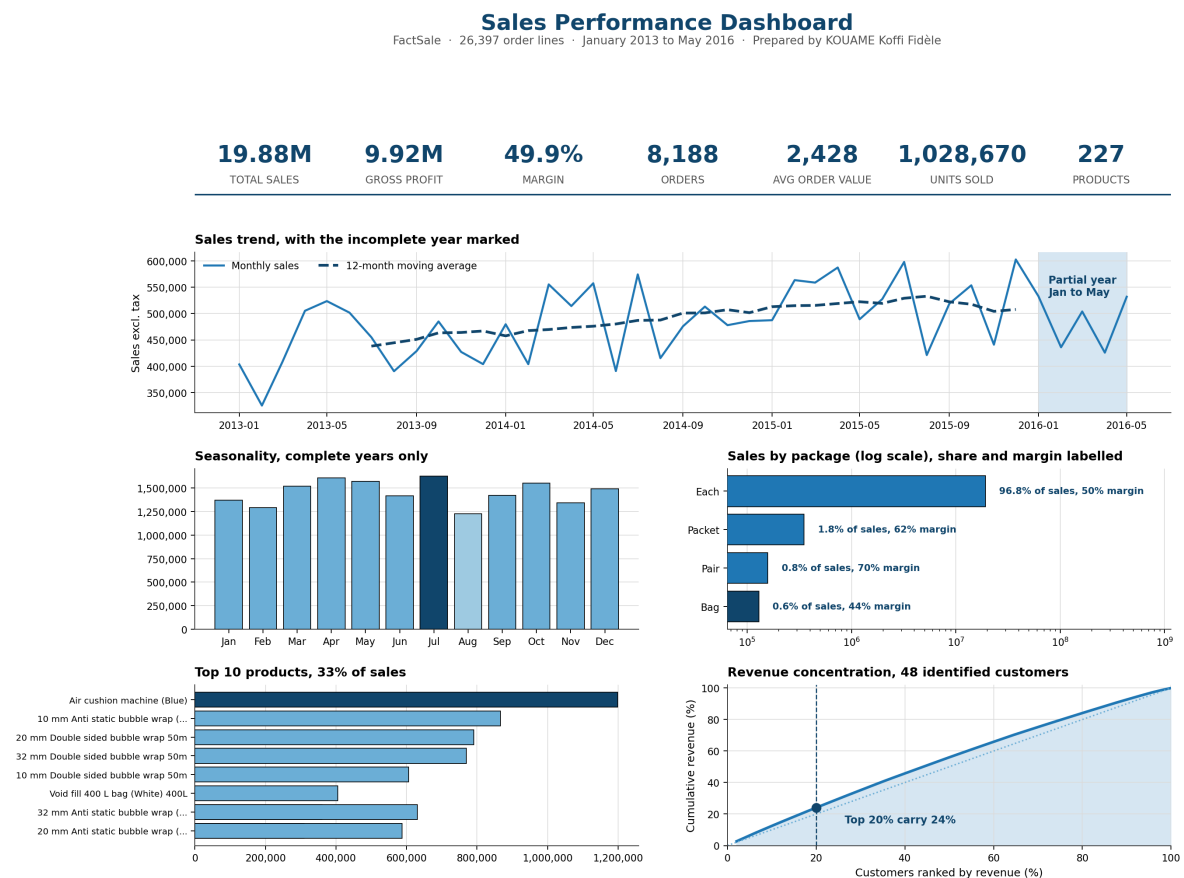


Figure 1. The sales performance dashboard. An interactive version, with hover detail, zoom and image export, is supplied as `dashboard.html` and opens in any browser without a server or an installation.

3.1 Why each chart type was chosen

Panel	Type	Reason
Sales trend	Line with moving average	Continuous time axis; the twelve-month window spans exactly one seasonal cycle, so the smoothed series isolates the trend
Seasonality	Vertical bars	Months are ordered categories; this answers which month sells most, which the time series cannot because each month appears several times
Package	Horizontal bars, log scale	One package type carries 96.8% of sales; on a linear axis the other three are invisible
Top products	Horizontal bars	Long product names read horizontally; only the top ten, since 227 bars answer nothing
Customer concentration	Cumulative curve	Concentration is a distributional property, which a ranking cannot show

A pie chart was deliberately not used. Comparing angles is less accurate than comparing lengths, and it fails entirely on a distribution as skewed as the package split.

4 Data quality

4.1 Checks that passed

Check	Result
Duplicate rows, full record	0
Duplicate business keys	0
Unparseable invoice dates	0
Quantity $\times$ Unit Price $\neq$ Total excl. tax	0
Total excl. $\times$ rate $\neq$ Tax amount	0
Total excl. + Tax $\neq$ Total incl.	0
Quantity ≤ 0	0
Negative sales	0
Delivery before invoice	0

4.2 Issues found and how each was handled

Unidentified customers. 9,077 lines carry a customer key of zero, which is a placeholder rather than a customer. The rows are **kept** for all revenue, product and time analysis, because their financial fields are complete and removing them would delete a third of the revenue. They are **excluded** from customer analysis only, through an explicit flag rather than a silent filter.

Incomplete final year. The file ends on 31 May 2016. Seasonality and every month-on-month comparison are computed on the complete years only, and the partial period is shaded in every time series so a reader cannot mistake it.

Missing delivery dates. 13 lines have none. They are **not imputed**: the field feeds a delivery-lag metric, and an invented date would manufacture a lag that never happened. The rows are retained, since every revenue field on them is intact.

Loss-making lines. 566 lines return a negative margin. These are **kept**. A negative margin on a real sale is a commercial event, not a data error, and removing it would inflate total profit.

High unit prices. 113 lines exceed a unit price of 1,000. Inspection shows every one is the same product, an air cushion machine at a consistent price, with plausible quantities and sound arithmetic. It is an expensive product, not a typing error, and it is **kept**.

4.3 The outlier trap

An interquartile screen at 1.5 times the range flags 2,337 lines on sales value, that is 8.9% of the file, and those lines carry **48.0% of all revenue**.

The screen was computed and **deliberately not applied**. The rule assumes an approximately symmetric distribution. Sales value is strongly right-skewed by construction, because a few large orders always exist. Applying a symmetric rule to a skewed variable does not remove errors; it removes the top of the distribution, which is precisely the revenue a sales analysis exists to explain.

5 Insights and recommendations

5.1 A third of revenue cannot be attributed to a customer

34.4% of order lines carry no customer key. This is a data capture failure upstream, and it is the most valuable finding of this audit: no segmentation, no lifetime value and no retention measure can be trusted until it is closed.

Recommendation. Trace the unattributed lines back to their source system and establish why the key is absent. Until then, publish every customer figure with its coverage rate beside it.

5.2 Products are concentrated, customers are not

The top ten products carry 32.8% of revenue out of 227 items. Among the 48 identified customers, however, the top 20% carry only 25% of identified revenue, against 20% under perfect equality.

Revenue is spread almost evenly across the account base, which is the opposite of the pattern most retail businesses show. There is no key-account dependency, so no single customer loss threatens the top line; but there is equally no high-value tier to build a retention programme around.

Recommendation. Do not build a key-account programme on this base, because the data says there are no key accounts. Concentrate commercial effort on the product side, and protect the availability and margin of the top ten items before anything else.

5.3 Price and volume move against each other, weakly

The correlation between unit price and quantity is -0.14, and between quantity and line revenue 0.24. The negative sign is expected: expensive items sell in smaller quantities. Its weakness matters more. Revenue per line is driven by price far more than by volume.

Recommendation. Do not use volume discounting as a revenue lever on this catalogue. The correlation structure says it will not recover the revenue it gives up.

5.4 Seasonality is real but moderate

On the complete years the strongest calendar month outsells the weakest by a factor of 1.33. That is worth planning around, but it does not justify large seasonal stock swings.

Recommendation. Align staffing and inventory to the quarterly pattern rather than the monthly one, whose amplitude does not warrant the operational cost of adjusting to it.

5.5 Loss-making lines are a control issue, not a strategic one

566 lines return a negative margin, costing 40,502. Against total profit that is 0.41%. Reporting it as a headline would be an error of proportion.

Recommendation. Add an automated flag on negative-margin lines at invoicing rather than opening a pricing review. The cost of the fix should match the size of the problem.

6 Limitations

The dataset contains no cost of acquisition and no marketing spend, so customer acquisition cost and lifetime value cannot be computed and are not reported. It contains no geographic

dimension beyond a city key that is not resolved to a place name, so no regional analysis is offered.

The figures are descriptive. No statistical test accompanies them. The differences discussed are large enough that this is unlikely to change the conclusions, but the point is stated rather than assumed.

Deliverables

File	Content
dashboard.html	Interactive dashboard, opens in any browser
report.pdf	The present report
Task2_Analysis.ipynb	Commented analysis notebook
DATA_QUALITY.md	Full quality audit
INSIGHTS.md	Insights and recommendations
data/FactSale_clean.csv	Cleaned dataset
charts/	Seven figures, PNG
results/	KPI and aggregate tables, plus the audit JSON

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